

Figure 2.4 Stars

exist today. Earlier Ptolemy, in his book *Almagest*, listed 48 constellations.

Ursa Major (Figure 2.3) is a constellation that can be seen in the northern hemisphere and part of the southern hemisphere. **Ursa Major** means **Great Bear** in Latin.

2.5 The Solar system

A solar system consists of a star (Figure 2.4) at the centre and the eight planets, moons, asteroids, comets and meteoroids that revolve it. The eight planets, namely the Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune, revolve around the sun in fixed elliptical paths known as '**orbits**'. Most stars host their own planets. So there are billions of other solar systems in the Milky Way galaxy alone.

Solar systems can also have more than one star. These are called binary star systems. If there are two stars or multi-star systems, if there are three or more stars. Our solar system is located in an outer spiral arm of the vast Milky Way galaxy. Our solar system orbits the centre of the Milky Way Galaxy at about 828,000 km/hr. Our solar system takes about 230 million years to complete one orbit around the galactic centre.

The solar system is believed to have been formed about 4.6 billion years ago. The solar system also includes the Kuiper Belt that lies past Neptune's orbit. This is a sparsely occupied ring of icy bodies. This is almost all smaller than the dwarf planet Pluto. Beyond the fringes of the Kuiper belt (Figure 2.5) is the Oort cloud. This giant spherical shell surrounds our solar system. It has never been directly observed,

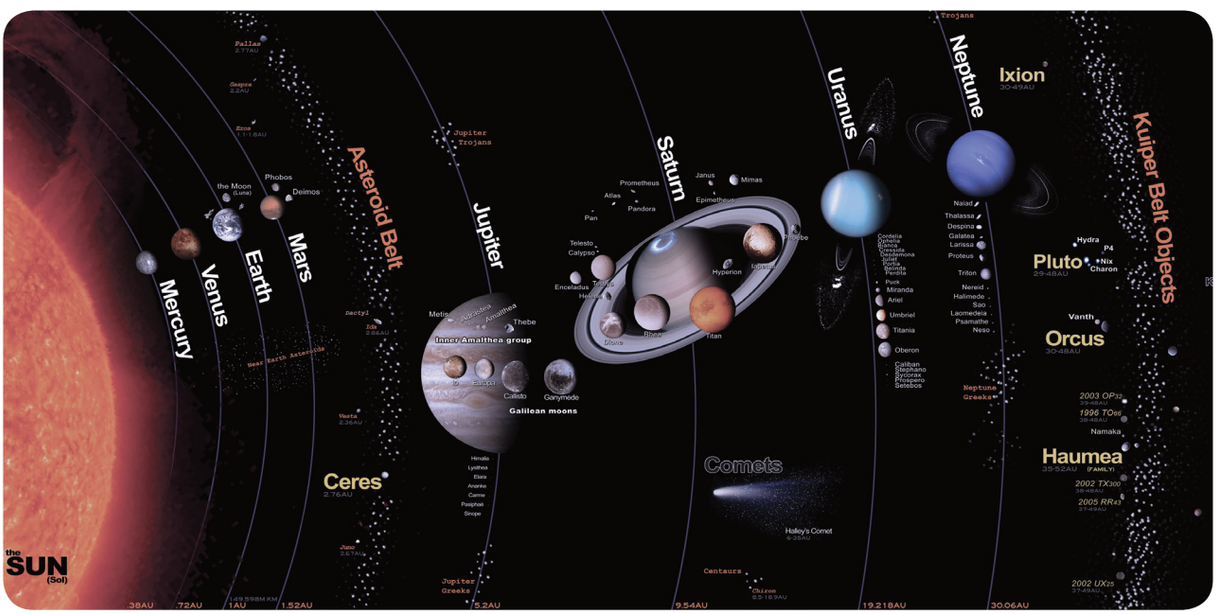


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